STRATEGIC ANALYSIS / 01

Decoding Customer Value

A SQL-driven retention strategy for a US D2C fashion brand. Prepared by Soumabha Bhim from the supplied customer snapshot.

Customers	Recorded sales	Promo sales share	Repeat cadence proxy
3,900	$233,081	42.65%	27.85%

The decision

The snapshot shows meaningful promotion exposure alongside an identifiable group of frequent, experienced customers. It does **not** demonstrate that promotions cause loyalty or that the business is dependent on discounts. Protect sales through a controlled offer-reduction pilot rather than a blanket withdrawal.

What the evidence supports

1. 1,677 customers (43.0%) have a promo flag; their recorded purchases total $99,411. These sales are associated with a promotion, not proven incremental sales.

2. Definition A identifies 1,086 customers with at least 26 previous purchases and stated cadence of at least monthly. Definition B identifies 817 with at least 26 previous purchases and a rating of at least 4.

3. High annual-spend-proxy customers are 20.0% of the base but account for only 25.7% of recorded sales. The data does not show an 80/20 concentration of observed revenue.

4. Arizona, Alaska and Tennessee lead a regularized, commercially screened geographic test shortlist. Acquisition efficiency remains unknown without market reach and CAC.

5. Start with 74 satisfied, established, high-proxy-value promo-exposed customers. Their recorded sales are $5,356; use them for pilot feasibility, not a powered claim of sales safety.

Evidence boundary

One row represents one customer and one recorded purchase. There are no timestamps, tenure duration, future outcomes, discount amounts, product costs or acquisition spend. All revenue figures refer to the supplied snapshot; no monthly or annual period can be assigned.

1 / Data preparation and quality

Audit	Result	Treatment
Rows / customer IDs	3,900 / 3,900 unique	All rows retained; no duplicates
Review ratings	37 missing (0.95%)	Preserve missing; satisfaction = Unknown
Discount vs promo code	Identical in all rows	Use one exposure signal; avoid double counting
Frequency labels	7 raw / 5 normalized	Bi-Weekly + Fortnightly = 26/year; Quarterly + Every 3 Months = 4/year
Gender / offers	No female promo customers	No female treated overlap; avoid causal gender claims
Subscriptions	All 1,053 subscribers promo exposed	No untreated subscriber counterfactual
Previous purchases	Range 1-50	No observed first-time customers; no one-time-buyer inference

Clean field names to snake_case, trim text, validate Yes/No domains, check IDs and numeric ranges. No arbitrary outlier removal or review imputation is applied. The raw CSV is retained unchanged and SHA-256 fingerprinted.

Granularity governs interpretation

State is available, but city and region are not. Previous purchases measure order history, not account tenure. Season is a category attached to the recorded purchase, not a dated seasonal time series. There is no valid RFM recency measure, churn model, two-year prediction or observed retention rate.

Selection and construction risks

Gender, subscription and promotion overlap sharply. Their relationships may reflect data construction or business eligibility rules; provenance is not supplied. Gender is not recommended as an acquisition exclusion. Validate actual offer eligibility before operational targeting.

2 / Engineered features and decisions

Feature	Definition	Decision / limitation
annual_orders_proxy	Weekly 52; two-weekly 26; monthly 12; quarterly 4; annual 1	Normalize stated cadence; plan contact frequency
dependency_score	100 x current promo flag (0 or 100)	Exposure screen only; historical dependency is unidentifiable
satisfaction_flag	Rating >=4 satisfied; <3.5 recovery; other known neutral; missing unknown	Protect dissatisfied / unknown customers from initial sunset
annual_spend_proxy_usd	Recorded amount x annual_orders_proxy	Value-tier prioritization; assumes stable order size and cadence
historical_value_proxy_usd	Recorded amount x (previous purchases + 1)	Rough history scale; not realized lifetime revenue
value_tier	High >=1924; Core >=572 and <1924; Developing <572	Fixed full-sample 80th / 50th percentile cutoffs
loyalty_a / loyalty_b	A: history >=26 and annual orders >=12; B: history >=26 and rating >=4	Competing definitions; A primary repeat-behavior proxy
history_band	0-10 / 11-25 / 26+ previous purchases	Compare category entry-like vs established mix
segment	Loyalty A yes/no x promo yes/no	Four traceable offer-treatment groups
sunset_eligible	Established promo exposed + High tier + rating >=4	Select controlled sunset pilot cohort

Age bands: 18-24, 25-34, 35-44, 45-54, 55-64, 65+. Other support fields: subscriber and satisfied binaries, high_history (26+), cadence_group, dependency_band, and value_tier_order. All definitions are implemented in src/pipeline.py and documented in the feature dictionary.

3 / Two loyalty definitions, tested

Test	A: history + cadence	B: history + satisfaction
Selected customers	1,086 (27.85%)	817 (20.95%)
Selected AOV	$59.58	$61.30
Other-customer AOV	$59.84	$59.36
AOV difference	$-0.26	$1.94
95% bootstrap CI	$-1.97 to $1.38	$0.16 to $3.84
Mean annual spend proxy	$1,733.51	$1,082.31
Satisfied share	42.91%	100% by construction
Definition overlap	Jaccard = 32.43%	Different constructs; not interchangeable

Primary choice: A, as an operational repeat-behavior proxy

A requires both accumulated purchase experience and ongoing stated cadence. It captures repeat behavior more directly than B, which can include annual shoppers with a favorable rating. The cadence-history conjunction supplies a clear operational logic for replenishment and relationship marketing. This is a choice of measurement construct, not evidence of future predictive accuracy.

Independent recorded spend does not validate A as a higher-ticket group: its AOV difference is near zero. B has a small exploratory positive difference, but satisfaction is 100% by definition and does not validate the label. Bootstrap intervals use 3,000 customer resamples per group, seed 42, assume independent customers, and are not adjusted for multiple comparisons.

Avoid circular validation

The annual spend proxy multiplies frequency, which is also part of A. Higher proxy spend in A is partly mechanical and must not be presented as independent proof. Value tiers and A also share cadence; dashboard scatter relationships inherit that construction.

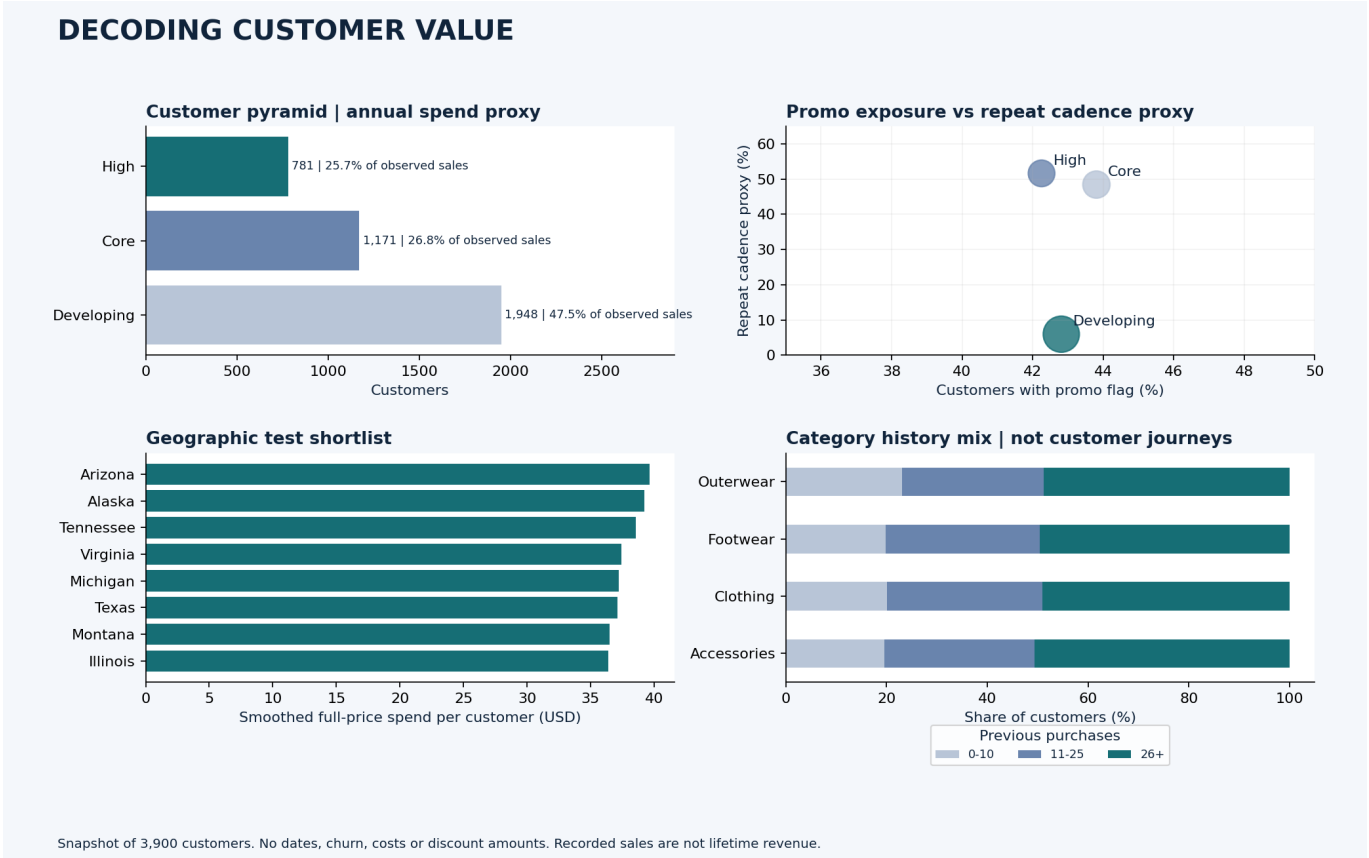
Robustness and future validation

threshold_sensitivity.csv evaluates 12 combinations: history thresholds 20, 26, 30, 38 and annual-order thresholds 4, 12, 26. These are operational sensitivity checks, not model tuning. Validate either definition prospectively against 90/180-day repurchase, contribution and survival outcomes once dates exist.

4 / Value distribution and customer segments

Tier	Customers	Recorded sales	Sales share
High	781	$59,879	25.7%
Core	1,171	$62,526	26.8%
Developing	1,948	$110,676	47.5%

Segment	Customers	Sales	Action
Developing full price	1614	$96,732	Improve second/next purchase journey
Developing promo exposed	1200	$71,645	Test smaller first incentives; do not label bargain hunters
Established full price	609	$36,938	Maintain full-price relationship
Established promo exposed	477	$27,766	Test targeted offer reduction



The dashboard uses value tiers for the promo/loyalty scatter. Using the four loyalty x promo segments there would produce deterministic 0/1 coordinates and provide little analytical information.

5 / Promotions and margin economics

Observed exposure: 43.0% of customers and 42.65% of recorded sales carry a promo flag. Discount amounts are missing, so actual discount cost, gross margin and incrementality cannot be calculated.

Within male non-subscribers, the promo vs full-price AOV difference is $-1.06 (95% bootstrap CI $-3.49 to $1.42). Restriction improves overlap but does not remove selection bias or establish treatment effect. Female shoppers have no promo observations and subscribers have no full-price observations.

Illustrative full-removal scenario: eligible cohort only

Assume the $5,356 recorded purchase total is net of a 15% discount, and COGS is 60% of list-price sales. These are planning assumptions, not facts in the dataset. The scenario rescales the cohort snapshot; it is not an annual profit forecast.

Volume loss	Scenario revenue	Gross profit	GP change
0%	$6,301	$2,520	+$945
5%	$5,986	$2,394	+$819
10%	$5,671	$2,268	+$693
20%	$5,041	$2,016	+$441

Baseline illustrative gross profit is $1,575. At 20% volume loss gross profit still rises, but revenue falls by $315: margin and sales protection are different constraints. The revenue-preserving loss ceiling is 15%; the gross-profit break-even loss is 37.5% under these assumptions.

Transparent equations

Let R = observed net sales; d = assumed discount; c = COGS/list sales; l = order-volume loss. List sales L = R/(1-d). Baseline GP = R-cL. Full-removal GP = (1-l)L(1-c). Revenue break-even loss = d. GP break-even loss = d/(1-c). Fixed costs, returns, shipping and offer-cost behavior are excluded.

margin_scenarios.csv contains 36 combinations of discount (10/15/20%), COGS (40/60/75%) and loss (0/5/10/20%). Partial discount reduction requires net multiplier (1-d_new)/(1-d_old), not the full-removal formula. Obtain order-level discount and cost data before setting a business case.

6 / Category and season evidence

Category	Customers	High-history share	AOV	Promo share
Accessories	1240	50.6%	$59.84	43.8%
Clothing	1737	49.1%	$60.03	42.1%
Footwear	599	49.6%	$60.26	43.2%
Outerwear	324	48.8%	$57.17	44.4%

High-history shares are close: 48.8%-50.6%. Outerwear has the largest low-history share (23.1%, versus about 19.5%-20.0% elsewhere) and the lowest AOV ($57.17). Treat this as a merchandising test hypothesis, not proof that outerwear attracts one-time shoppers.

Seasonal mix

High-history shares are Fall 47.6%, Spring 47.9%, Summer 51.9%, Winter 51.1%. These small cross-sectional differences do not establish a seasonal retention pattern. Exposure, category availability and customer mix are unobserved.

Answers to the category funnel request

The four-panel dashboard compares each category across low (0-10), mid (11-25) and high (26+) purchase history. Because each customer has one item/category and history is total brand purchases, the chart does not show the first category bought, category-level repeat purchases or transitions. Entry-point and retention categories remain proxies.

Commercial action and trade-off

Test outerwear follow-up recommendations after purchase and maintain broad clothing/accessory coverage. Compare personalized cross-sell with a no-message holdout. Track 90-day contribution per eligible customer, conversion, unsubscribe and returns. The risk is wasted contacts or unwanted bundles; do not shift inventory materially on these small descriptive gaps.

All query outputs are included: category_funnel, season_patterns and q2_behavior. Revenue figures and proportions reconcile to the customer-level dataset.

7 / Geographic opportunity and demographics

State	n	AOV	Promo	Smoothed full-price $/customer
Arizona	65	$66.55	33.8%	$39.65
Alaska	72	$67.60	40.3%	$39.25
Tennessee	77	$61.97	36.4%	$38.56
Virginia	77	$62.88	37.7%	$37.41
Michigan	73	$62.10	39.7%	$37.20
Texas	77	$61.19	36.4%	$37.15
Montana	96	$60.25	37.5%	$36.52
Illinois	92	$61.05	40.2%	$36.39

Screen: at least 60 customers, AOV above $59.76, and promo customer share below 43%. Rank by (state full-price sales + 50 x national full-price sales/customer)/(state customers + 50). The 50-customer prior reduces sensitivity to small samples; it is a heuristic, not an estimated causal effect.

Kansas ranks third on regularized full-price spend but fails the AOV screen, demonstrating why rank and commercial qualification are separate. Arizona, Alaska and Tennessee are the top qualified candidates; no state is proven underpenetrated without population, impressions, media spend and market share.

State-level targeting action

Run matched-budget, randomized audience or geo tests in Arizona and Tennessee; use Alaska as a separate test because shipping economics could differ. Primary outcome: incremental contribution after CAC, returns and delivery cost. Trade-off: lower promo exposure may reflect customer selection, and remote shipping can eliminate a higher basket-value advantage.

Demographic boundaries

Age has almost no linear association with recorded purchase amount (r=-0.010); previous purchase count has r=0.008 with purchase amount. Do not create a narrow age target based on weak aggregate signals. Preserve age/gender slices for descriptive monitoring, and use behavior and product affinity for acquisition tests.

The SQL geographic and demographic views provide auditable candidate screens. They cannot estimate commercial underleveraging because market opportunity denominators are absent.

8 / Ideal customer profile and acquisition use

Operational seed: value tier High, loyalty A=1, promo_exposed=0 and review_rating>=4. This yields 103 customers (2.64% of the base). These are satisfied, experienced, frequent customers with a full-price purchase observed; persistent full-price loyalty is not established.

Profile dimension	Evidence	How to use
Cadence	55 weekly; 48 every two weeks	Choose replenishment reminders; frequency is partly selected by rule
Age	25-34: 24.3% vs 19.4% base	Test broad adjacent ages; no age exclusion
Category	Clothing 43.7%; footwear 23.3% vs 15.4% base	Test footwear creative and clothing cross-sell
Payment	Credit card 24.3% vs 17.2% base	Keep convenient checkout; preference is not a causal value driver
Satisfaction	All >=4 by cohort rule	Use service quality as a safeguard, not external validation
Gender	Female 54.4% vs 32.0% base	Do not target/exclude by this: promo overlap mechanically biases the seed

Acquisition brief

Reach shoppers with demonstrated apparel/footwear interest using full-price product-quality creative, fit guidance and convenient checkout. Test modeled audiences based on the seed only where consent and platform policy allow; exclude current customers from prospecting evaluation. Compare with broad-interest acquisition using equal budgets and a holdout.

Success and trade-off

Track incremental contribution per acquired customer, new-customer CAC, 90-day repeat purchase and return rate. Risk: a small, mechanically selected seed can overfit and exclude good occasional shoppers. Keep 20% of test budget for broad exploration; do not project seed frequency into future CLV.

What the ideal profile is not

It is not proof that a particular age, state, payment method or gender causes loyalty. Rankings from a 103-customer seed are descriptive. Per-level counts, shares and population indices are supplied in icp_profile.csv.

9 / Reproduction, verification and next data

Reproducible deliverables

Python: raw data, cleaned/engineered CSV, all summary outputs, reproducible scripts and an executable notebook. SQL: nine views plus a populated SQLite database. Power BI: four-panel PBIP project with embedded source data, model, report definitions, DAX measures and theme. A standalone offline HTML companion provides filters, a schematic state map and CSV export.

Verification performed

Validated 3,900 unique customer rows, domains and ranges; preserved 37 unknown ratings; reconciled SQL customer counts and recorded sales; verified loyalty rules and embedded Power BI data against CSV; generated all SQL outputs and deterministic bootstrap results. PDFs are rendered for visual checks. Power BI Desktop runtime validation is separate from source/schema checks; see powerbi/OPEN_IN_POWER_BI.md for status and steps.

Minimum data contract for genuine retention

Required field / table	Decision unlocked
Order ID, customer ID, order timestamp, signup date	Cohorts, recency, tenure and future repurchase
List price, net price, discount amount, offer ID	Actual discount depth and promotional cost
Randomized offer assignment and exposure log	Incremental effect and discount response
SKU-level order lines, returns, COGS, fulfillment cost	Category transitions and contribution margin
Acquisition source, campaign spend, exposure, geography	Incremental CAC and geographic opportunity
Consent, communications, service contacts	Responsible targeting and service recovery

Sources and provenance

Business requirements: supplied Problem Statements1.pdf, three pages. Analysis source: supplied Dataset.csv, 3,900 rows and 18 fields. No external customer data has been substituted. Microsoft documentation was used only for the Power BI file format, not for business findings.

Power BI format references: https://learn.microsoft.com/en-us/power-bi/developer/projects/projects-report ; https://learn.microsoft.com/en-us/power-bi/developer/projects/projects-dataset . Raw-data SHA-256 is recorded in data/processed/metrics.json.